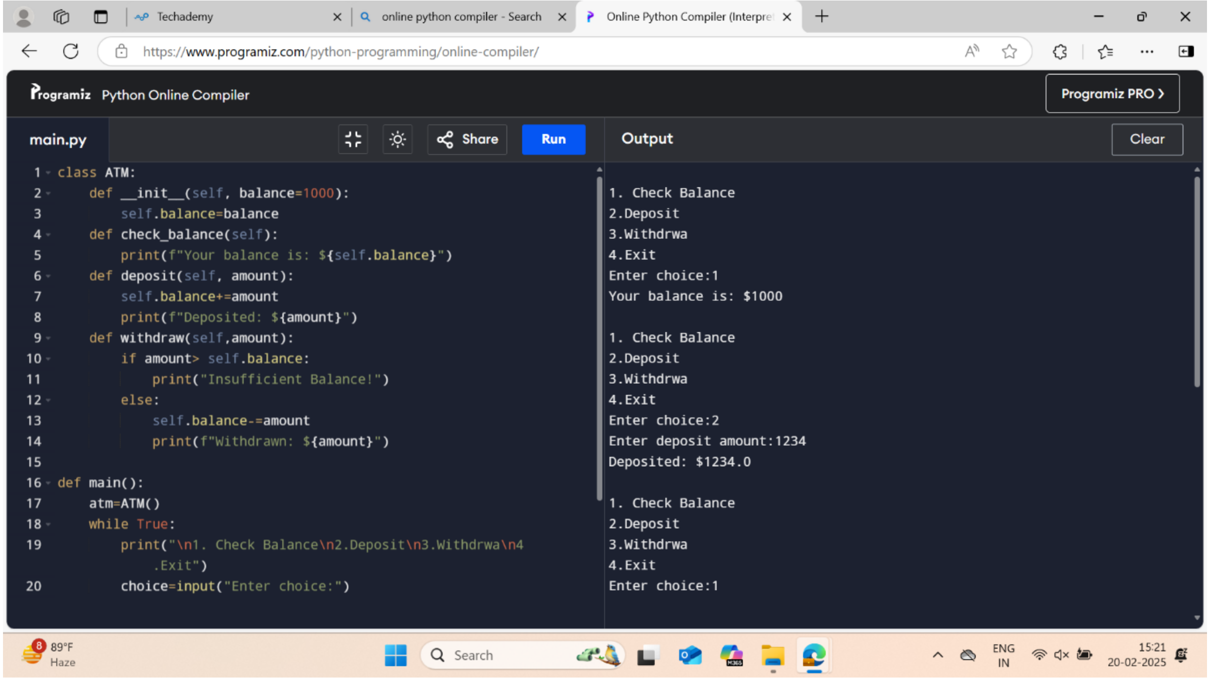
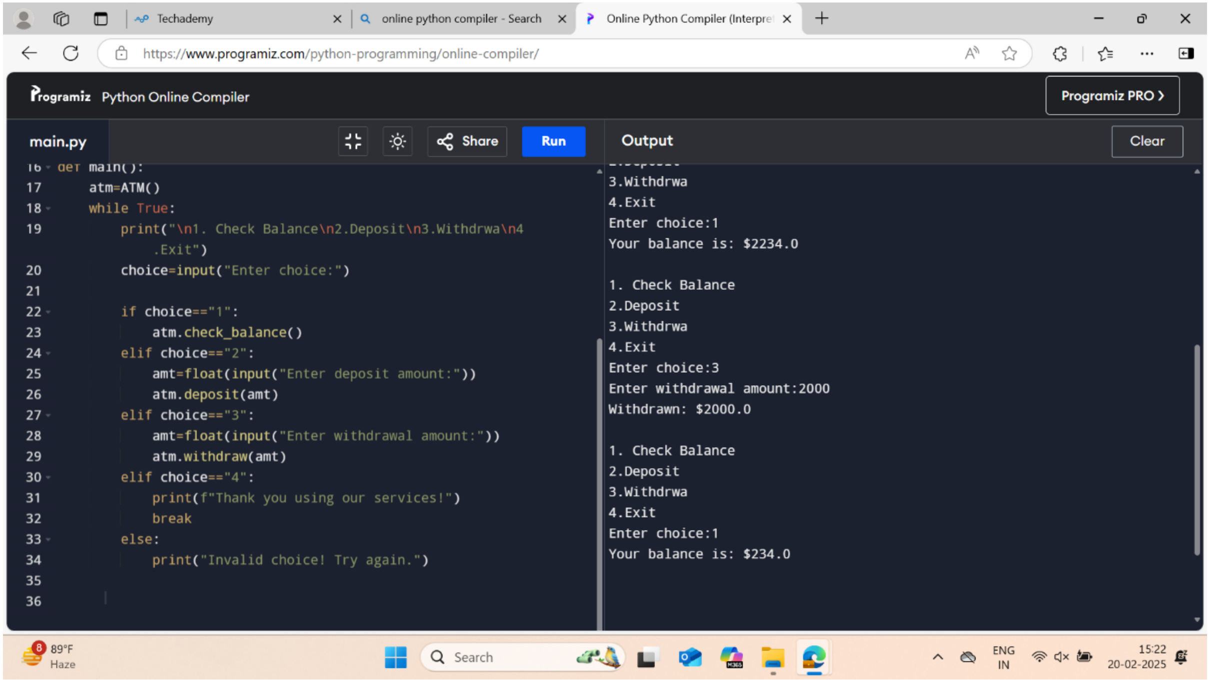
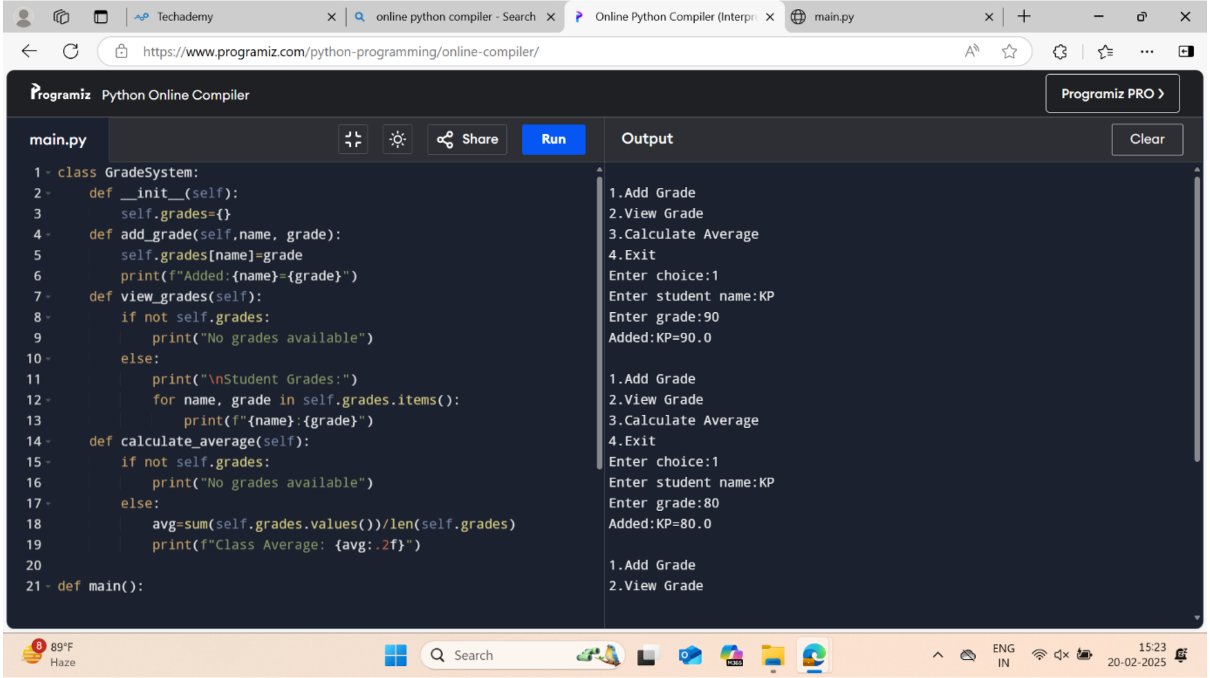
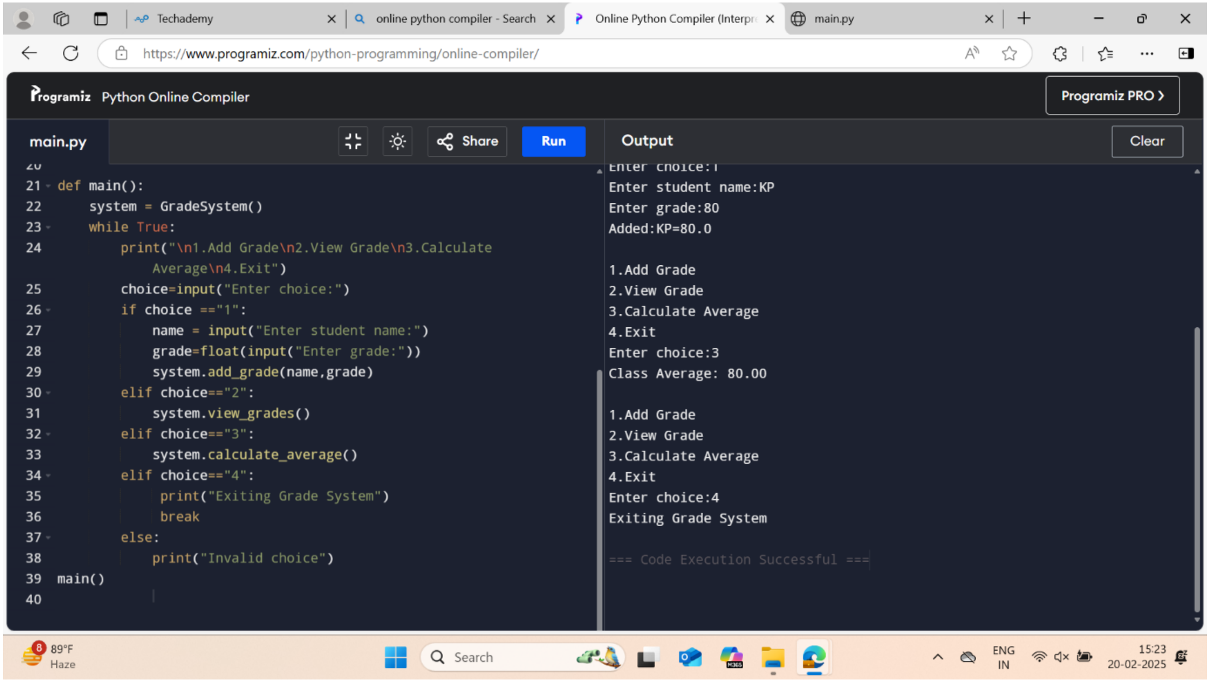
Q. ATM Simulation





Q. Student Grading System





Q. Hospital Management System

main.py

```python
class Hospital:
    def __init__(self):
        self.patients={}
    def add_patient(self,id,name,age,disease):
        self.patients[id]={"Name":name,"Age":age,"Disease":disease}
        print(f"Patient {name} added!")
    def view_patients(self):
        if not self.patients:
            print("No patients registered!")
        else:
            print("\nPatient Records:")
            for id,details in self.patients.items():
                print(f"ID: {id} - {details}")
    def remove_patient(self,id):
        if id in self.patients:
            del self.patients[id]
            print("Patient removed")
        else:
            print("Patient not found!")
def main():
```

Output

```
1.Add Patient
2.View Patients
3.Remove Patient
4.Exit
Enter choice:1
Enter Patient ID:12AB
Enter Name:KP
Enter Age:22
Enter Disease:Sore Throat
Patient KP added!

1.Add Patient
2.View Patients
3.Remove Patient
4.Exit
Enter choice:2

Patient Records:
ID: 12AB - {'Name': 'KP', 'Age': '22', 'Disease': 'Sore Throat'}
```

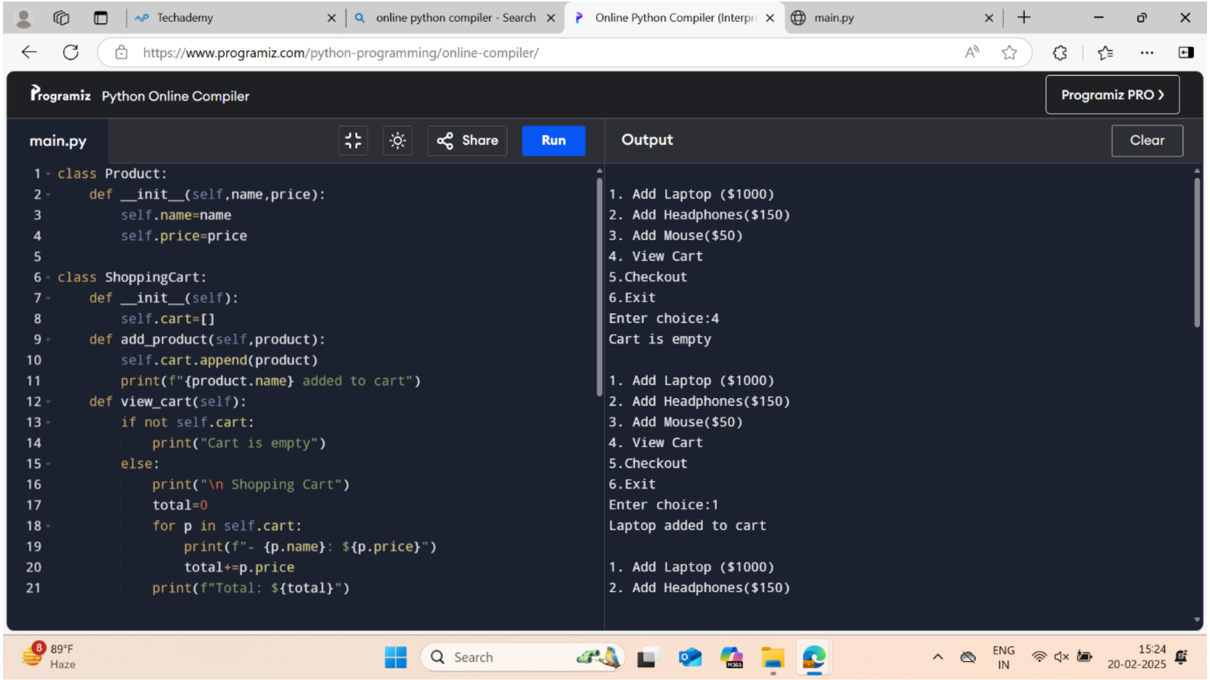
main.py

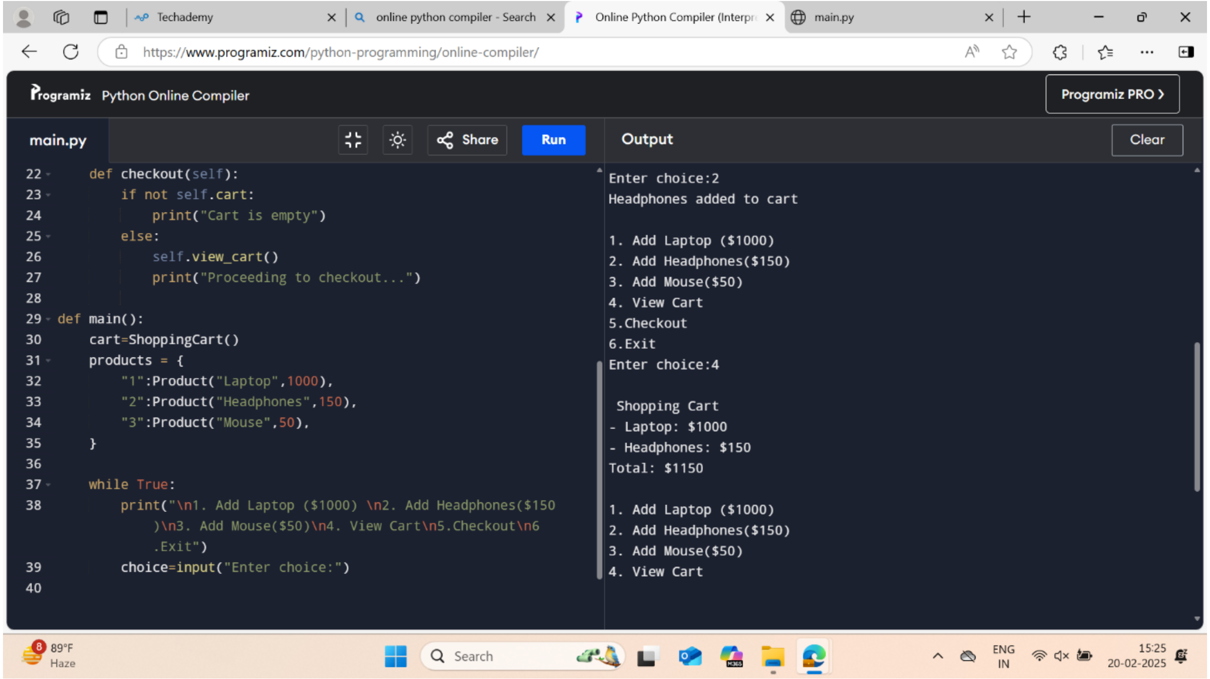
```python
def main():
    hospital =Hospital()
    while True:
        print("\n1.Add Patient\n2.View Patients\n3.Remove Patient\n4.Exit")
        choice=input("Enter choice:")
        if choice =="1":
            id=input("Enter Patient ID:")
            name=input("Enter Name:")
            age=input("Enter Age:")
            disease=input("Enter Disease:")
            hospital.add_patient(id,name,age,disease)
        elif choice=="2":
            hospital.view_patients()
        elif choice == "3":
            id=input("Enter Patient ID to remove:")
            hospital.remove_patient(id)
        elif choice == "4":
            print("Exiting Hospital System")
            break
        else:
```

Output

```
1.Add Patient
2.View Patients
3.Remove Patient
4.Exit
Enter choice:3
Enter Patient ID to remove:12AB
Patient removed

1.Add Patient
2.View Patients
3.Remove Patient
4.Exit
Enter choice:2
No patients registered!

1.Add Patient
2.View Patients
3.Remove Patient
4.Exit
Enter choice:4
```

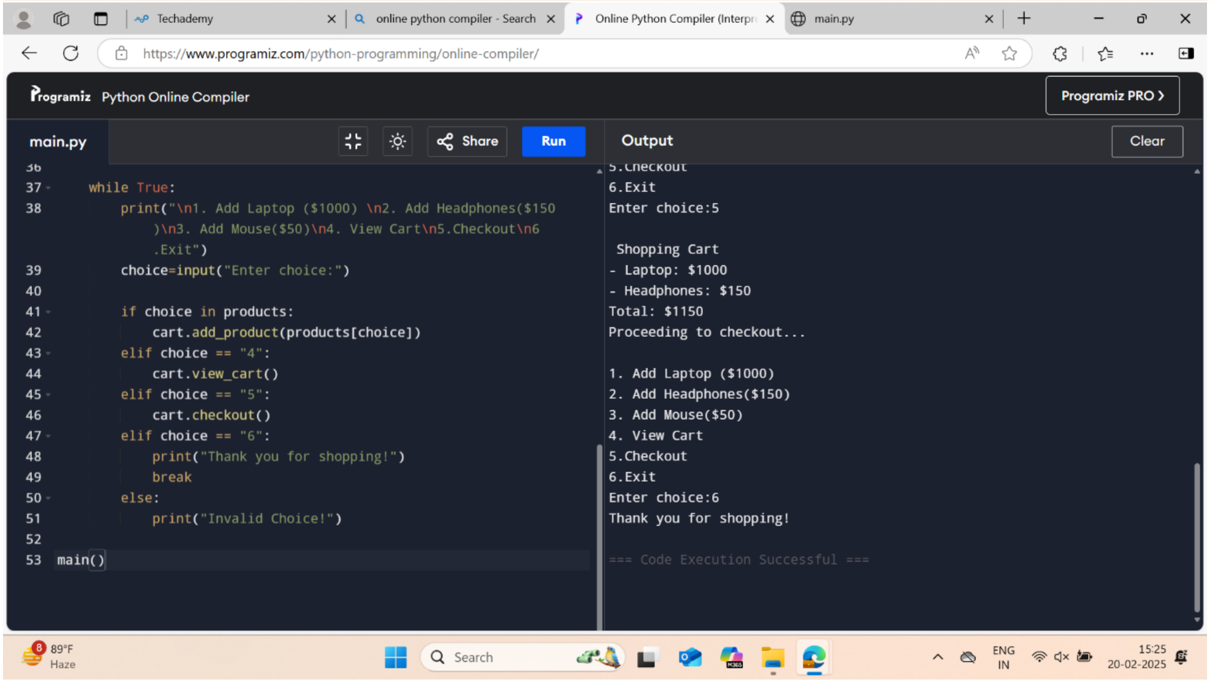
Q. Order Management System






```python
    def checkout(self):
        if not self.cart:
            print("Cart is empty")
        else:
            self.view_cart()
            print("Proceeding to checkout...")

def main():
    cart=ShoppingCart()
    products = {
        "1":Product("Laptop",1000),
        "2":Product("Headphones",150),
        "3":Product("Mouse",50),
    }

    while True:
        print("\n1. Add Laptop ($1000) \n2. Add Headphones($150
            )\n3. Add Mouse($50)\n4. View Cart\n5.Checkout\n6
            .Exit")
        choice=input("Enter choice:")

```

```
Enter choice:2
Headphones added to cart

1. Add Laptop ($1000)
2. Add Headphones($150)
3. Add Mouse($50)
4. View Cart
5.Checkout
6.Exit
Enter choice:4

 Shopping Cart
- Laptop: $1000
- Headphones: $150
Total: $1150

1. Add Laptop ($1000)
2. Add Headphones($150)
3. Add Mouse($50)
4. View Cart
```


```python

    while True:
        print("\n1. Add Laptop ($1000) \n2. Add Headphones($150
            )\n3. Add Mouse($50)\n4. View Cart\n5.Checkout\n6
            .Exit")
        choice=input("Enter choice:")

        if choice in products:
            cart.add_product(products[choice])
        elif choice == "4":
            cart.view_cart()
        elif choice == "5":
            cart.checkout()
        elif choice == "6":
            print("Thank you for shopping!")
            break
        else:
            print("Invalid Choice!")

main()
```

```
5.Checkout
6.Exit
Enter choice:5

 Shopping Cart
- Laptop: $1000
- Headphones: $150
Total: $1150
Proceeding to checkout...

1. Add Laptop ($1000)
2. Add Headphones($150)
3. Add Mouse($50)
4. View Cart
5.Checkout
6.Exit
Enter choice:6
Thank you for shopping!

=== Code Execution Successful ===
```